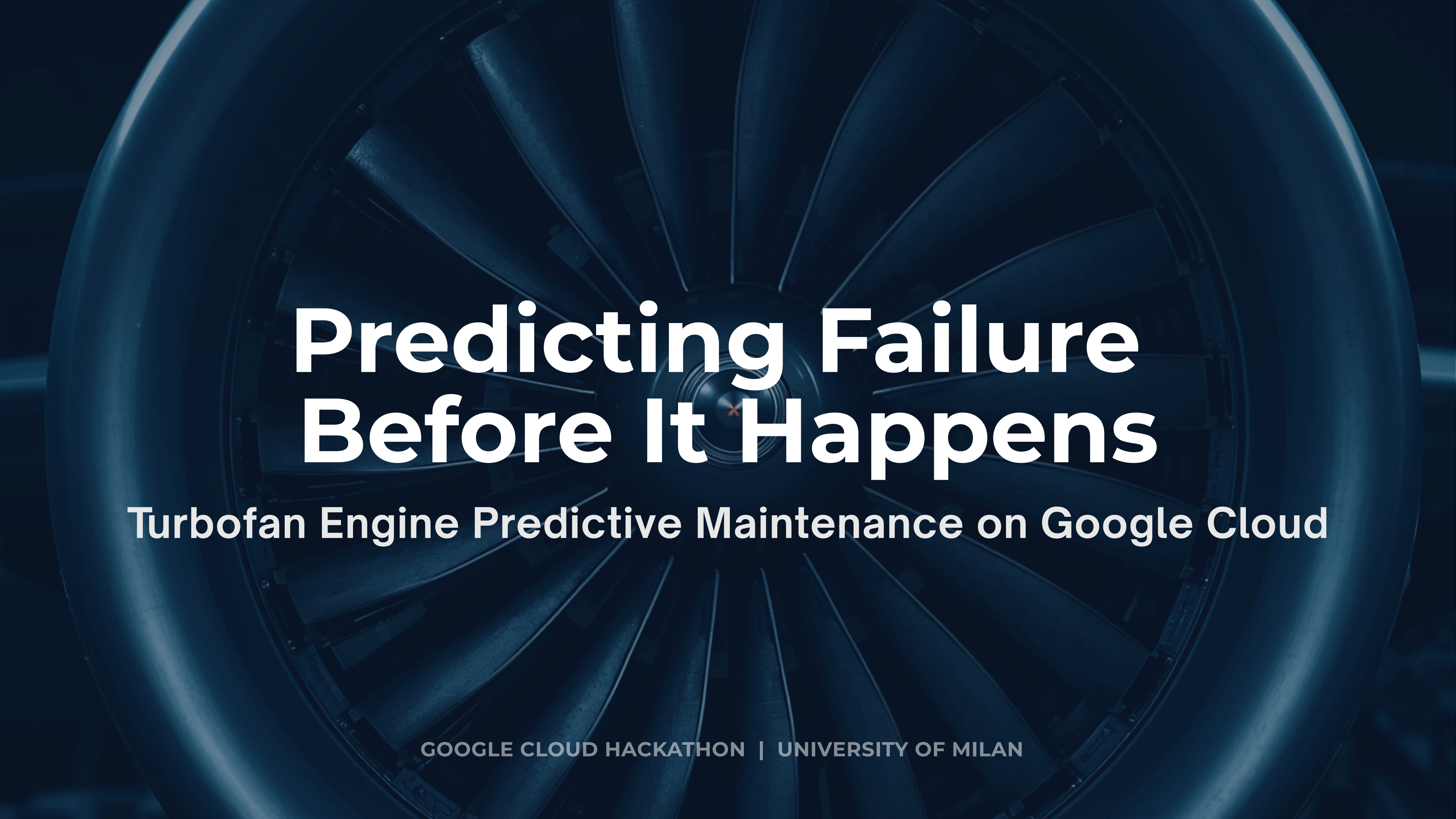




Predicting Failure Before It Happens

Turbofan Engine Predictive Maintenance on Google Cloud

GOOGLE CLOUD HACKATHON | UNIVERSITY OF MILAN

The Problem

Why predictive maintenance matters

\$10K–\$30K/hr

Cost of AOG event per hour

35–40%

of airline operating costs
is engine maintenance

2–5x

more expensive
(unplanned vs scheduled
maintenance)

"airBaltic cancelled 4,670 flights due to engine issues. Wizz Air reported a 61.7% drop in operating profit from grounded aircraft." — Air Cargo Week, 2026

Dataset & Model

The Data

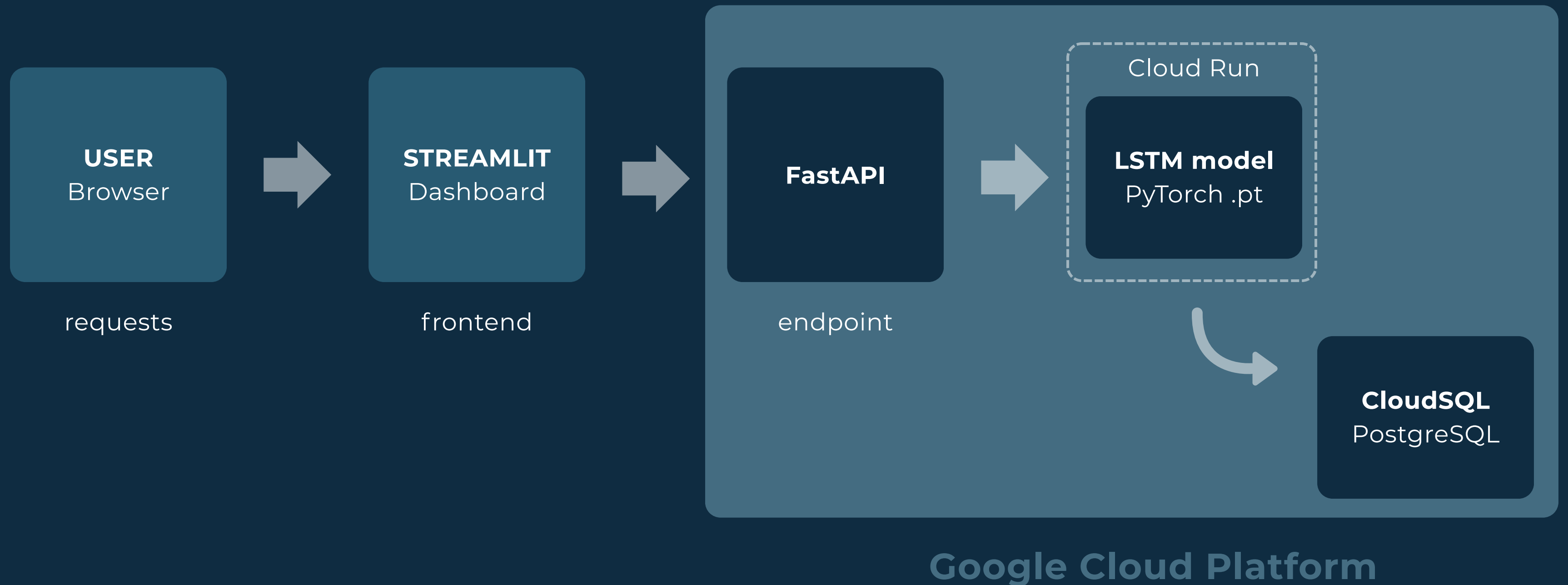
- NASA C-MAPSS dataset
- 100 turbofan engines
- 21 sensors per engine
- Each cycle = one flight

The Model

- LSTM in PyTorch
- Predicts remaining flights to failure
- Test RMSE: 12.81 cycles

Unlike classical models, LSTM learns how engines degrade over time, not just their current state.

Deployment Architecture



Live Demo

Turbofan engine health monitor deployed on google cloud run

→ PREDICTIVE MAINTENANCE · GCP

Turbofan Engine Health Monitor

NASA C-MAPSS dataset · LSTM model · Deployed on Google Cloud Run

TOTAL ENGINES

92

across fleet

 CRITICAL

44

immediate action

 WARNING

12

schedule soon

 MONITOR

18

keep an eye on

 HEALTHY

18

operating normally

Monthly Cost Estimate

Google Cloud Platform · Production scenario · 10,000 requests/month

Cloud Run

Serverless container

FastAPI + LSTM

\$0.10 / month

Cloud SQL

PostgreSQL

db-f1-micro

10GB

\$9.37 / month

Cloud Storage

Model artifacts (.pt, .pkl)

1GB

\$0.00 / month

Total:

\$9.47 / month

Estimate from Google Cloud Pricing Calculator · May 2026